

LinGraph workflow

Locus graph preparation

Whole-genome assemblies

1

Identify novel loci

Non-redundant large insertions as novel loci
Improve downstream annotation

OPTIONAL

2

Partition whole genomes into loci

Define locus-based blocks

3

Lift loci using k-mers

Map corresponding loci across assemblies

4

Build and align locus graphs

Construct local graphs and align assemblies

5

Pair query and reference sequences

Use k-mers for sequence liftover and pairing

6

Linearize graph-anchored alignments

Generate pairwise sequence alignments

7

Fill gaps between loci

Align intervening sequences

8

Call grVCF and merge cohorts

Generate sample calls and merge cohort variants

9

Build the pangenome graph

Convert grVCF to GFA

Alignment and cohort outputs

Skip optional step

